

Project: Sales Forecasting and Demand Prediction

🌀 What is the problem we are trying to solve?

Many businesses struggle with accurately predicting how much of a product they will need in the future. Without reliable sales and demand forecasts, companies often face challenges like overstocking, running out of products, or missing opportunities to increase revenue. These issues not only affect day-to-day operations but can also lead to long-term financial losses.

This project focuses on developing a forecasting system that uses historical sales data to better anticipate future demand. The goal is to build a system that helps businesses make **informed, data-driven decisions** about:

- How much inventory to stock.
- When to launch promotions.
- Where to allocate resources across stores or regions.
- How to optimize sales strategies for revenue growth

🌀 Data Exploration:

We began by exploring the dataset, which included historical records of product sales, store locations, discount rates, and several other business-related variables. The initial analysis helped us understand the distribution of data, identify missing values, detect outliers, and uncover patterns such as seasonal trends or relationships between variables like discounts and sales volume. Interactive visualizations were also created to offer dynamic views of key metrics and patterns across time and different stores or product categories.

🌀 Feature Engineering:

To improve model performance, we performed feature engineering by creating new variables that capture hidden patterns in the data. These included time-based features like month, day of the week, and holiday indicators, as well as derived metrics like lagged sales values, rolling averages, and discount ratios. These additional features helped the model better understand the underlying trends and seasonality in customer behavior.

🌀 Model Development:

For the forecasting model, we experimented with several machine learning algorithms such as Linear Regression, Random Forest, and Gradient Boosting. Each model was evaluated based on accuracy metrics such as RMSE (Root Mean Squared Error) and MAE (Mean Absolute Error). After tuning hyperparameters and comparing performance, the best-performing model was selected to make future sales predictions. The results were integrated into a dashboard to allow end users to interact with the forecasts and visualize the effects of variables like discounts on sales.

➤ *Throughout this project, we set out to understand and predict sales and demand patterns in a way that could genuinely help businesses make smarter decisions. By carefully exploring the data, engineering meaningful features, and developing accurate forecasting models, we were able to build a system that does more than just make predictions, it tells a story about what drives customer behavior.*

This work isn't just about numbers. It's about giving businesses the tools to avoid overstocking or running out of products, to prepare better for seasonal changes, and to respond more confidently to market shifts. While there's always room to grow, like integrating live data or automating updates, we believe this project is a strong step toward smarter, data-backed planning.

About Dataset

The dataset used is the "Global Super Store Dataset" available on Kaggle. This dataset contains historical sales data for a global superstore, providing insights into consumer behavior and product performance across different regions. It is well-suited for time-series analysis and forecasting due to its detailed transactional data. The Global Super Store Dataset contains approximately 51,291 entries and 24 columns.

Column Descriptions:

1. Row ID: Row identifier.
2. Order ID: Order identifier.
3. Order Date: Date of order.
4. Ship Date: Date of shipment.
5. Ship Mode: Shipping method.
6. Customer ID: Customer identifier.
7. Customer Name: Customer's name.
8. Segment: Customer type.
9. City: City of order.
10. State: State of order.
11. Country: Country of order.
12. Postal Code: Postal code.
13. Market: Market region.
14. Region: Geographical region.
15. Product ID: Product identifier.
16. Category: Product category.
17. Sub-Category: Product subcategory.
18. Product Name: Product name.
19. Sales: Sales amount.
20. Quantity: Number of items.
21. Discount: Discount applied.
22. Profit: Profit earned.
23. Shipping Cost: Shipping cost.
24. Order Priority: Order priority level.

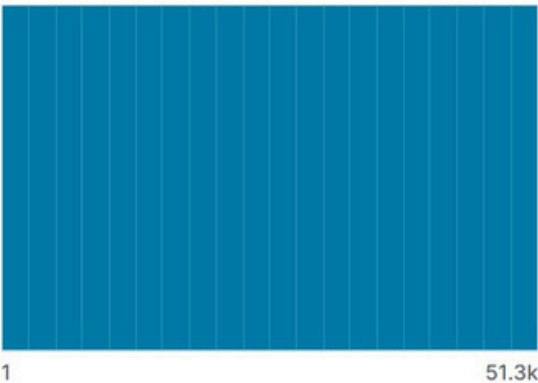
Dataset Characteristics:

The dataset is diverse, containing both categorical and numerical data. It includes temporal information with "Order Date" and "Ship Date" in datetime format. Some columns like "Cost Price," "Retail Price," and others related to monetary values are currently stored as objects, which may need conversion for accurate numerical analysis. The dataset provides a comprehensive snapshot of the sales process, making it suitable for various analytical and exploratory tasks.

A sample of the Data:

40155	CA-2014-114-10-2014	21-10-2014	Standard CJW-15220	Jane Waco	Corporate	Sacrament	California
40936	CA-2012-128-01-2012	31-01-2012	Second ClaJH-15985	Joseph Ho	Consumer	Concord	North Caro
34577	CA-2011-15/4/2011	9/4/2011	Second ClaGM-14695	Greg Maxw	Corporate	Alexandria	Virginia
28879	ID-2012-2819-04-2012	22-04-2012	First ClassAJ-10780	Anthony Ja	Corporate	Kabul	Kabul
45794	SA-2011-1827-12-2011	29-12-2011	Second ClaMM-7260	Magdelene	Consumer	Jizan	Jizan
4132	MX-2012-113-11-2012	13-11-2012	Same DayVF-21715	Vicky Freym	Home Offi	Toledo	Parana
27704	IN-2013-736/6/2013	8/6/2013	Second ClaPF-19120	Peter Fulle	Consumer	Mudanjian	Heilongjian
13779	ES-2014-5031-07-2014	3/8/2014	Second ClaBP-11185	Ben Peterm	Corporate	Paris	Ile-de-Fran
36178	CA-2014-1#####		Second ClaTB-21175	Thomas Bo	Corporate	Henderson	Kentucky
12069	ES-2014-168/9/2014	14-09-2014	Standard CPJ-18835	Patrick Jon	Corporate	Prato	Tuscany
22096	IN-2014-1131-01-2014	1/2/2014	First ClassJS-15685	Jim Sink	Corporate	Townsville	Queenslan
49463	TZ-2014-81#####		Second ClaRH-9555	Ritsa Hight	Consumer	Uvinza	Kigoma
46630	PL-2012-788/8/2012		First ClassAB-600	Ann Blume	Corporate	Bytom	Silesia

🔗 Row ID



Valid	51.3k	100%
Mismatched	0	0%
Missing	0	0%
Mean	25.6k	
Std. Deviation	14.8k	
Quantiles		
	1	Min
	12.8k	25%
	25.6k	50%
	38.5k	75%
	51.3k	Max

⚙ Order ID

25035
unique values

Valid	51.3k	100%
Mismatched	0	0%
Missing	0	0%
Unique	25.0k	
Most Common	CA-2014-1...	0%

⚙ Order Date

1430
unique values

Valid	51.3k	100%
Mismatched	0	0%
Missing	0	0%
Unique	1430	
Most Common	18-06-2014	0%

A more detailed information about some of the columns in the data (Missing values, unique values and most common)

Data Collection, Exploration, and Preprocessing:

Objective:

The goal of this phase was to prepare a clean and structured dataset that is ready for building forecasting models.

What We Did:

1. Data Collection:

- Loaded a dataset containing historical sales data for multiple stores and items.

```
In [58]: sales = pd.read_csv('../Data/Original Data.csv', encoding='latin-1')
sales.shape
```

Out[58]: (51290, 24)

 Distribute Dataset

- The data included columns 24 columns.

```
In [59]: sales.describe()
```

Out[59]:

	Row ID	Postal Code	Sales	Quantity	Discount	Profit	Shipping Cost
count	51290.00000	9994.000000	51290.000000	51290.000000	51290.000000	51290.000000	51290.000000
mean	25645.50000	55190.379428	246.490581	3.476545	0.142908	28.610982	26.375915
std	14806.29199	32063.693350	487.565361	2.278766	0.212280	174.340972	57.296804
min	1.00000	1040.000000	0.444000	1.000000	0.000000	-6599.978000	0.000000
25%	12823.25000	23223.000000	30.758625	2.000000	0.000000	0.000000	2.610000
50%	25645.50000	56430.500000	85.053000	3.000000	0.000000	9.240000	7.790000
75%	38467.75000	90008.000000	251.053200	5.000000	0.200000	36.810000	24.450000
max	51290.00000	99301.000000	22638.480000	14.000000	0.850000	8399.976000	933.570000

- Then we dropped useless columns like 'Row ID', 'Order ID', 'Customer ID', 'Postal Code', 'Product ID'
- Then added some features like the holidays using USFederalHolidayCalendar

2. Data Exploration:

Investigate relationships between product types, promotional activities, and sales volume. In addition to understanding sales trends, seasonality, and external factors influencing demand.

- Reviewed the structure and content of the dataset.

Like getting the highest selling category

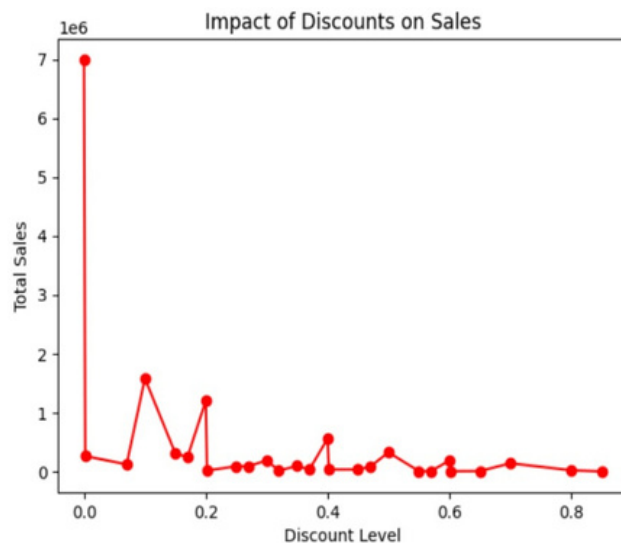
In [68]:

```
category_sales = sales.groupby('Category')['Sales'].sum().round(2).sort_values(ascending=False)
category_sales = pd.DataFrame(category_sales).reset_index()
category_sales.index += 1
category_sales
```

Out[68]:

	Category	Sales
1	Technology	4744557.50
2	Furniture	4110874.19
3	Office Supplies	3787070.23

The highest selling category is Technology



total sales initially drop sharply with a small increase in discount level but then fluctuate and gradually decline



as sales increase, profit generally goes up too, but there is some scatter, and sometimes higher sales can still lead to a loss.

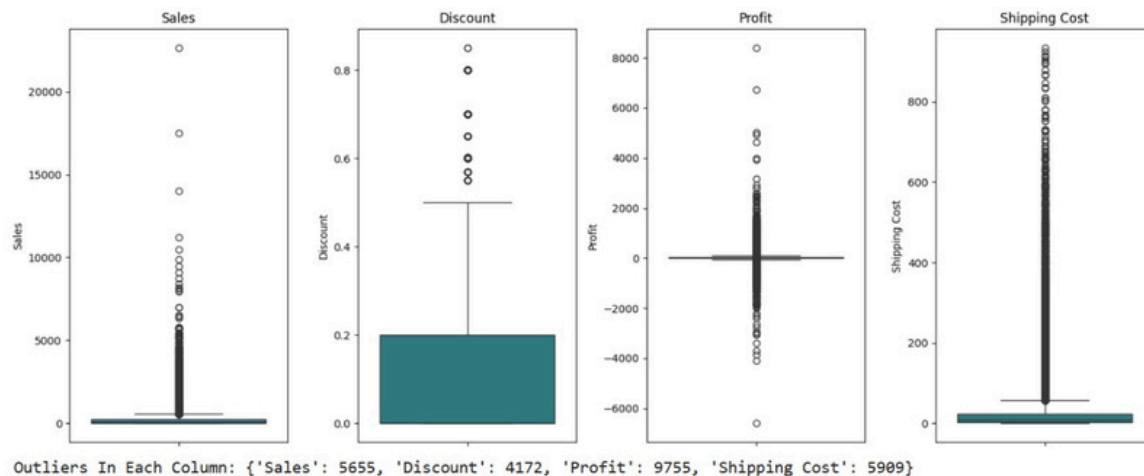
- Then we started Handlingmissingvalues,duplicates,andoutliers, and compute basic summary statistics.

1-The data did not have any missing values or any duplicates.

```
missing_values = pd.DataFrame(sales.isna().sum())
missing_values
```

2-Identifying the outliers

By Plotting the Outliers in each numerical column and calculating the IQR



The sales and shipping costs have some extreme outliers, discounts are mostly consistent with some high values, and profits vary widely with some significant negative outliers.

- Visualized general sales trends over time to understand seasonality and growth patterns.

3. Data Preprocessing:

- Converted the date column to a proper datetime format.

```
# Convert Order Date to DateTime
sales['Order Date'] = pd.to_datetime(sales['Order Date'])

# Convert Ship Date to Date time
sales['Ship Date'] = pd.to_datetime(sales['Ship Date'])
```

- Added new time-related features like year, month, and day to support better analysis.
- Verified data consistency and ensured there were no missing values or extreme outliers that could affect modeling.

4. Feature Engineering:

- Created time-based features such as month, day, week and adding the season to help capture patterns in the sales.